# Understanding Large Language Models' Ability on Interdisciplinary Research

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## Abstract

Recent advancements in Large Language Models (LLMs) have revealed their impressive ability to perform multi-step, logic-driven reasoning across complex domains, positioning them as powerful tools and collaborators in scientific discovery while challenging the long-held view that inspiration-driven ideation is uniquely human. However, the lack of a dedicated benchmark that evaluates LLMs' ability to develop ideas in Interdisciplinary Research (IDR) settings poses a critical barrier to fully understanding their strengths and limitations. To address this gap, we introduce IDRBench — a pioneering benchmark featuring an expert annotated dataset and a suite of tasks tailored to evaluate LLMs' capabilities in proposing valuable research ideas from different scientific domains for interdisciplinary research. This benchmark aims to provide a systematic framework for assessing LLM performance in complex, cross-domain scientific research. Our dataset consists of scientific publications sourced from the ArXiv platform covering six distinct disciplines, and is annotated by domain experts with diverse academic backgrounds. To ensure high-quality annotations, we emphasize clearly defined dimensions that characterize authentic interdisciplinary research. The design of evaluation tasks in IDRBench follows a progressive, real-world perspective, reflecting the natural stages of interdisciplinary research development, including *1) IDR Paper Identification*, *2) IDR Idea Integration*, and *3) IDR Idea Recommendation*. Using IDRBench, we construct baselines across 10 LLMs and observe that despite fostering some level of IDR awareness, LLMs still struggle to produce quality IDR ideas. These findings could not only spark new research directions, but also help to develop next-generation LLMs that excel in interdisciplinary research.

**Data & Code:** https://anonymous.4open.science/r/IDRBench-Framework/

**Dataset Card:** https://huggingface.co/datasets/IDRBench/IDRBench/

## 1 Introduction

*"If you would understand anything, observe its beginning and its development."*
*—Aristotle*

Aristotle, one of history's greatest polymaths, made enduring contributions across philosophy, biology, physics, logic, and more areas, shaping centuries of intellectual development. However, as the body of scientific knowledge has grown considerably, modern science has fragmented into ever more

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specialized fields, making it increasingly difficult to bridge interdisciplinary scientific domains. Related to this, recent advancements in Large Language Models (LLMs) [27, 26, 11, 1, 9], with access to massive corpora spanning diverse scientific fields and improved reasoning abilities, have shown enormous potential in scientific discovery. Frameworks that endow LLMs with nascent abilities to conduct research ideation [31, 24], deploy research codes [3], or even compose detailed research proposals [32] have catalyzed the emergence of a handful of benchmarks measuring LLM's ideation quality [12, 20, 23, 38], marking a promising new paradigm of LLM-accelerated scientific research.

While these frameworks and benchmarks provide solid evaluations on the outcome quality of the proposed ideas in generic research or single-disciplines, there is a notable gap in evaluating LLM's ability to perform ideation when the research context is *Interdisciplinary Research (IDR)*. Existing works that explicitly explore the potential of using LLMs to tackle IDR problems [21, 36, 22, 39] often conduct investigations that provide different approaches and frameworks, yet lack a thorough evaluation of the inherent LLM capability to perform IDR tasks. As a result, the excessive focus on outcome-driven analyses of LLMs tends to overlook IDR's core characteristic of **integration of knowledge across distinct disciplines**, leaving scarce evidence to answer this key research question: ***Are LLMs capable of conducting interdisciplinary research?***

Answering this question is not easy. Specifically, we need to evaluate whether an LLM can capture the "Eureka" moment in interdisciplinary research – a flash of sudden enlightenment arising from different disciplines during which a new scientific discovery is made [16]. However, to the best of our knowledge, none of the existing works have assessed such capacity of LLMs. One major issue that hinders such evaluation is the lack of a dedicated benchmark specifically designed for interdisciplinary research. The absence of benchmark also makes it more difficult to measure the actual improvements brought by new approaches when they are compared to the baselines, increasing the risk of misleading comparisons and reducing the reproducibility of IDR findings.

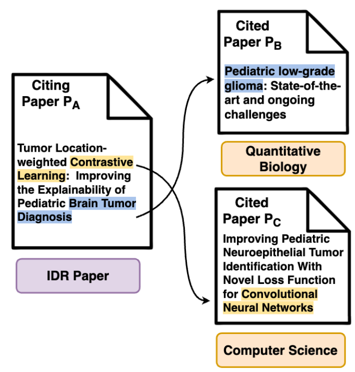


Figure 1: Triplet data format in IDRBench - Showing that Papers $P_B$ and $P_C$ are integrated (more than merely referenced) to generate the IDR Paper $P_A$. To obtain a positive triplet, the annotators need to identify $P_A$ from the candidate pool, then figure out the key cited papers $P_B$ and $P_C$ that play central roles in deriving the IDR idea.

To contribute to understanding LLMs' ability in IDR, we introduce **IDRBench**, a pioneering benchmark designed for IDR assessment. IDRBench adopts the perspective of a progressive assessment of LLMs conducting interdisciplinary research by introducing three tasks, including *(i) IDR Paper Identification (IPI)*, evaluating LLM's ability to classify interdisciplinary papers; *(ii) IDR Idea Integration (I3)*, testing LLM's ability to combine papers from distinct disciplines into feasible and novel IDR; and *(iii) IDR Idea Recommendation (I2R)*, evaluating LLM's ability to recommend the most appropriate IDR paper given a list of unrelated papers.

To gauge LLMs' performance on these tasks, the IDRBench dataset is structured as a knowledge triplet as illustrated in Figure 1. For instance, an IDR paper titled "Tumor Location-weighted Contrastive Learning: Improving the Explainability of Pediatric Brain Tumor Diagnosis" serves as "Citing Paper $P_A$", accompanied by two "Cited Papers" $P_B$ and $P_C$ from the disciplines of quantitative biology and computer science, respectively. In this way, an IDR triplet is represented in the form $[P_A; (P_B, P_C)]$.

A comprehensive evaluation using IDRBench on 10 mainstream LLMs establishes the first quantified baseline that accurately measures the inherent capability of various LLMs in conducting IDR. Our results also show divergent performance on each of the three IDR tasks, with LLMs demonstrating an advantage over both IDR Idea Integration (I3) and IDR Idea Recommendation (I2R) tasks, while leaving room for improvement in the IDR Paper Identification (IPI) task [3]. Further analysis on these results reveals that although LLMs have developed some awareness of the IDR notions in IDR classification and

[3]Our progressive assessment is derived from the real world scenario of human researchers conducting IDR. However, our experiments in Section 5 indicates interesting observations that LLMs do not typically follow this progressive setup.

| Dataset | IDR | Progressive Evaluation | Source | Data Type | # Papers | Data Format |
|---|---|---|---|---|---|---|
| **IDRBench (Ours)** | ✓ | ✓ | ArXiv | Annotated [4] + Synthetic | 9,485 | Structured paper triplet $[P_A; (P_B, P_C)]$ |
| SchNovel[20] | ✗ | ✗ | ArXiv | Synthetic | 15,000 | Paper similarity score |
| IdeaBench[12] | ✗ | ✗ | Semantic Scholar | Synthetic | 2,374 | Biomedical paper + Reference list |
| Research-Bench[23] | ✗ | ✓ | Semantic Scholar + CrossRef | Annotated + Synthetic | 1,386 | Paper + Reference list |

Table 1: Comparison of IDRBench and existing general ideation benchmarks. For each positive paper triplet in IDRBench, the annotators are asked to complete 4 tasks, including identifying the IDR paper $P_A$ and at least 2 cited papers $P_B$ and $P_C$ that formulates the idea of $P_A$.

recommendation, they are still struggling to yield quality IDR ideas in a plug-and-play fashion under real-world settings, leaving space for future improvements.

We summarize our main contributions as follows:

- We release IDRBench, the first ever benchmark with human annotation to comprehensively evaluate LLMs' ability in the Interdisciplinary Research (IDR) setting, including a human-annotated dataset and a suite of dedicated tasks, enabling a thorough assessment of LLM's IDR ability.
- To authentically reflect LLMs strengths and limitations in realistic IDR environments, we evaluate LLM performance under a closed-book setting on three progressive tasks, including: IDR Paper Identification (IPI), IDR Idea Integration (I3), and IDR Idea Recommendation (I2R).
- Our evaluation and analysis results reveal divergent performance across LLMs, showcasing their advantages in both I3 and I2R tasks, while leaving notable room for improvement in the IPI task. Further analysis also suggests that LLMs are still struggling to provide high-quality IDR proposals despite showing early signs of IDR awareness.

## 2 Related Works

**Human-LLM Collaboration in IDR.** Recent works explore the potential of LLMs in interdisciplinary research mostly from a human-LLM collaborative perspective. For instance, [39] proposes Disciplink, a novel framework that assists researchers in seeking and gathering potential IDR topics based on their domains. Similarly, [22] leverages LLM persona simulation with human-in-the-loop integration to enhance IDR topic formulation. Other attempts focus on specific tasks within the IDR. A handful of techniques, including summary text-generation [7, 18], sample-efficient learning [10], and content-aware analysis [33] are utilized in various frameworks [36, 21] to conduct IDR paper classification. However, the majority of the existing frameworks demonstrate performance gains via modules that enhance LLMs externally, overlooking the inherent capability of these models in IDR.

**Benchmarking LLM's Research Ideation Ability.** Motivated by existing LLM-assisted research ideation frameworks [29, 32], several studies propose novel benchmarks that aim to evaluate the quality of research ideas. For example, [20] introduces SchNovel dataset along with a retrieval-augmented framework to assess the novelty of proposed research ideas. [12, 23, 38, 37] take the perspective of hypothesis generation to curate large-scale datasets that benchmark the performance of LLMs in formulating novel yet feasible hypotheses. Table 1 summarizes the key features of IDRBench when compared to other existing benchmarks. To the best of our knowledge, IDRBench is the first to explicitly provide an evaluation on interdisciplinary research (as shown in the IDR column). Moreover, the second column refers to benchmarks that incorporate progressive evaluation with increasing complexity, while the Source column indicates the origin of each dataset. Different from other benchmarks, we also introduce the paper triplet shown in the Data Format column to emphasize multiple discipline integration as well as allowing human annotation. Finally, IDRBench offers a decent scale in total paper quantity when compared to other benchmarks.

[4] Experts were paid to annotate the positive paper triplets at an hourly rate higher than the minimum wage in the location where the annotation and research were conducted.

# 3 The IDRBench

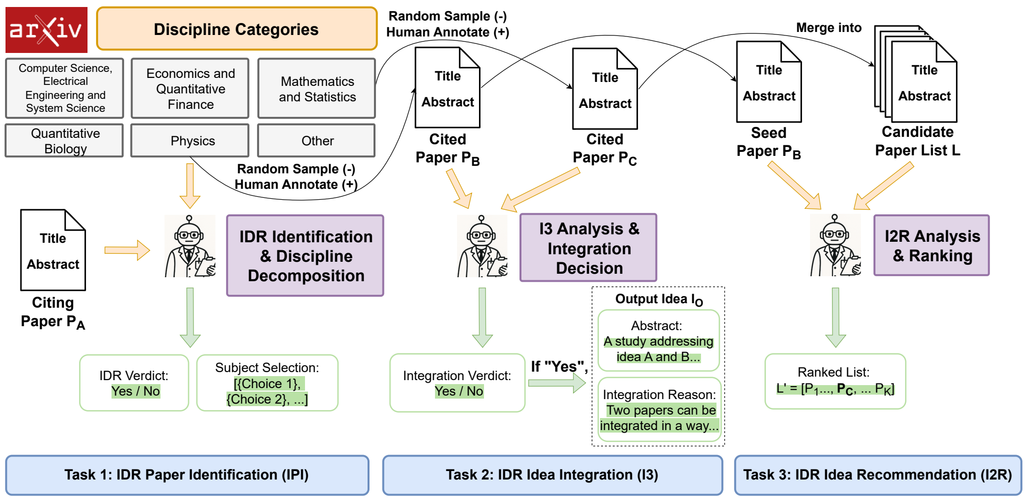


Figure 2: Visualization of tasks IPI, I3, and I2R within IDRBench. Orange and green arrows stand for input and output flow, respectively. Purple boxes summarize the LLM's functions.

We adopt the widely used definition of "Interdisciplinary Research" (IDR) [6, 15, 25, 30]: *"a mode of research by teams or individuals that integrates information, data, techniques, tools, perspectives, concepts, and/or theories from two or more disciplines or bodies of specialized knowledge to advance fundamental understanding or to solve problems whose solutions are beyond the scope of a single discipline or area of research practice."* Following the IDR definition to model the integration of ideas from disciplines that are contextually distant, our proposed IDRBench provides a novel benchmark of two key components: (i) a dataset based on paper triplets, and (ii) a suite of three progressive evaluation tasks.

## 3.1 The IDRBench Dataset

Our dataset in IDRBench employs the knowledge triplet $[P_A; (P_B, P_C)]$ showcased in Figure 1, which was used in our annotation process and is explored in different ways in our proposed tasks (in Section 3.2). From the triplet, the citing paper $P_A$ is an interdisciplinary paper that cites and integrates cited papers $P_B$ and $P_C$ from distinct disciplines. To form this triplet, we derive a subset from the ArXiv Platform[5] with over 58,444 papers between November 2024 and January 2025.

Our choice of using ArXiv platform to ensure a more accurate evaluation of LLMs' IDR capabilities stems from two main reasons: (i) many of the recent papers uploaded have not yet been published in any formal venue, which reduces the risk of information leakage during LLM pretraining (also noted by [23]), and (ii) ArXiv requires authors to select specific category taxonomies (referred to as disciplines in our work) when submitting papers. Moreover, we select specific combinations of disciplines that are conceptually more distant from one another, e.g., Computer Science + Quantitative Biology. We also merge some closely related disciplines into single categories to reflect their conceptual proximity, e.g., Computer Science and Electrical Engineering.

For positive instances, we collected 120 instances of structured triplets containing 360 papers. To obtain one positive triplet, the human expert needs to go through the reference list of $P_A$ (with the average list length of 35 papers) to pick out $P_B$ and $P_C$. This selection is important, as it describes the main representative papers of each discipline being combined in $P_A$. We also performed a double-check agreement within the positive samples across experts to guarantee more reliable results. Further details regarding the annotation process are provided in Appendix A.1.

For negative instances, we perform a random selection of 9,125 from the same ArXiv subset (out of 58,444 papers) to create structured paper triplets, with stratified sampling strategies tailored to

[5]https://arxiv.org/

the goal of each task. The rational of applying randomness for negatives is two-folded: (i) [4] notes that interdisciplinary connections are less frequent and more sparse compared to non-IDR papers (ii) our discipline analysis among all papers in ArXiv (over 2,282,079 papers) finds that more than 85% of papers are marked by the authors as mono-discipline, which further enhances the findings in [4]. Details of the negative sampling method are discussed in Section 3.2.

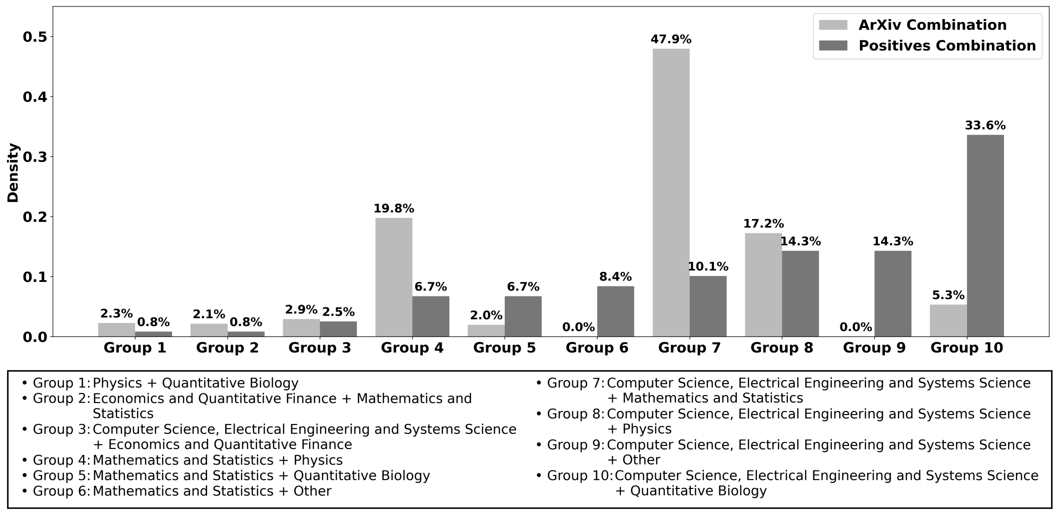


Figure 3: Distribution of binary discipline combinations for ArXiv data and positive samples in IDRBench. There are 15,099 samples in ArXiv and 120 samples in IDRBench.

Figure 3 provides a comparison between our dataset's discipline distribution expressed in binary combinations and the discipline distribution of ArXiv taxonomy. As we can see, the portion of more distant discipline combinations is notably higher in the "Positive Combination" of our dataset compared to that of the "ArXiv Combination". "Positive Combination" refers to the paper evaluated by our annotators (our positive cases) while 'ArXiv Combination" refers to the description of all papers in ArXiv. For example, very diverse Groups 8 and 10 (which combine Computer Science with Physics and Quantitative Biology, respectively) represent nearly 50% of our annotated dataset, while they account for only 23% of the ArXiv distribution. Moreover, a less diverse group such as Group 7 (Mathematics + Computer Science) constitutes only 10% of our dataset compared to 47% in the ArXiv. Finally, we use "Others" to refer to disciplines identified by the annotators that are not included in the ArXiv taxonomy. As a result, our dataset features more diverse groups than ArXiv with enhanced interdisciplinary representativeness.

### 3.2 The IDRBench Tasks

Figure 2 provides an overview of the tasks defined in IDRBench, which include *(i) IDR Paper Identification (IPI)*, *(ii) IDR Idea Integration (I3)*, and *(iii) IDR Idea Recommendation (I2R)*, evaluating different stages of conduct interdisciplinary research.

#### 3.2.1 IDR Paper Identification (IPI)

**IPI Task Description.** The pipeline of **I**DR **P**aper **I**dentification (IPI), shown in the first column in Figure 2, measures the fundamental ability of LLM to identify whether an scientific work is an IDR. Given the title and abstract of a citing paper $P_A$, LLM is first instructed to perform classification on whether $P_A$ is an IDR from at least two distinct disciplines. In case of a positive classification, it is further instructed to refer to the discipline categories derived from ArXiv taxonomies (within the top left corner in Figure 2) and decompose the IDR idea in $P_A$ into a combination of discipline choices.

**Dataset in IPI.** We build the IPI dataset with negative and positive examples, where each instance is a paper $P_A$ description (paper and title), a label (yes or no), and (in the case of yes) two disciplines integrated in $P_A$. The positive instances are derived from human annotation results, considering 120 papers. For negative ones, we select from ArXiv papers where authors only defined one single discipline in their submission. In fact, although these papers may offer novel contributions, their scope is confined to a single discipline with no interdisciplinary overlap. Following our observation in Section 3.1, we set an approximate 1:10 ratio between annotated positive and curated negative samples to reflect the nature of IDR research and ensure a realistic evaluation setup.

#### 3.2.2 IDR Idea Integration (I3)

**Task Description.** **I**DR **I**dea **I**ntegration (I3), showcased in the second column in Figure 2, takes a step further to investigate LLMs IDR integration ability. We provide LLMs with two cited papers $P_B$ and $P_C$ that contain the title and abstract. LLM is thereafter instructed to perform an idea integration analysis under an IDR setting, predicting whether the integration between $P_B$ and $P_C$ is a promising IDR. The integration analysis focuses on three criteria, including (1) whether the potential outcome idea $I_o$ aligns with the official definition of IDR, (2) whether the integration of the two papers yields a feasible IDR, and (3) whether the output IDR idea bears sufficient novelty. We assume that the combination of $P_B$ and $P_C$ is only viable when all three criteria are met. If the answer is yes, we further prompt the model to generate an IDR idea $I_o$ represented by a working paper abstract and the integration reasoning using $P_B$ and $P_C$. The reasoning output facilitates our understanding and provides informative insights during the idea integration stage of IDR.

To sufficiently evaluate whether LLMs are truly identifying IDR, one need to ensure that papers $P_B$ and $P_C$ exhibit a meaningful and reliable distance between their disciplines. To address this challenge, we construct two subsets with different difficulty levels.

**Dataset 1 in I3 - Level 1.** For positive pairs, we use the cited papers $(P_B, P_C)$ from our annotation process, considering that both were used to be integrated in the IDR $P_A$ paper. Our dataset has 120 pairs of positive samples. For the negative pairs in difficulty level 1, two papers are randomly selected from two distinct disciplines. This approach is based on the observation that valid interdisciplinary ideas are extremely sparse within the overall space of possible paper combinations. Therefore, this strategy considers negativity from the undirected aspect of randomness, i.e., as long as two random papers are selected from two random yet distinct disciplines, they are considered a negative pair. In dataset 1, we also use a 1:10 ratio between annotated positive and curated negative pairs.

**Dataset 2 in I3 - Level 2.** For positive pairs, we follow Level 1 settings. However, for negative ones, we introduce a more challenging semi-random selection method in difficulty. We now consider that two papers randomly selected from **the same discipline** are negative instances. Thus, from a positive pair $P_B$ and $P_C$, we replace $P_B$ with randomly selected paper $P_D$ from the same discipline as $P_C$. These negative pairs are considered more difficult, as paper $P_D$ can be related to $P_C$ since both are in the same discipline but may not necessarily form a valid IDR idea. For Dataset 2, as there is a combination of $(P_B, P_D)$ and $(P_C, P_D)$ as negative instances, we maintain an approximate 1:20 ratio between annotated positive and curated negative pairs.

#### 3.2.3 IDR Idea Recommendation (I2R)

**Task Description.** **I**DR **I**dea **R**ecommendation (I2R) is the most complex task in IDRBench, since it tests the LLMs' holistic IDR ability in a multiple-step recommendation setting. Given a positive cited paper pair $P_B$ and $P_C$, the LLM receives $P_B$ as the seed paper, while the other paper $P_C$ is merged into a candidate paper list $L = \{P_1, P_2, ...P_k\}$ with size $K$, that contains unrelated papers. The model is then instructed to summarize ideas from seed $P_B$, search through the useful ideas in $L$, and produce a re-ranked paper list $L'$, following similar criteria described in I3, considering novelty, feasibility, and IDR. In I2R, the model is asked to make a comparison to investigate the feasibility of conducting promising IDR when each paper $P_i \in L$ is paired with $P_B$. This task performs better when $P_C$ is at the top of $L'$ after LLM's re-ranking. Following previous works where ranking is typically performed via pairwise comparison to ensure reliability and robustness [32], we apply a similar setting using the ranking module shown in the middle of the third column in Figure 2.

Similar to the I3 task, we also propose two different datasets with stratified difficulty levels to perform evaluation. In both datasets, an instance of the dataset consists of a seed paper $P_B$, a list $L$ with $K$ negative papers, and one positive paper $P_C$, forming a recommendation task.

**Dataset 1 in I2R - Level 1.** For level 1, the list $L$ contains a set of random negative papers. Specifically, we have $P_B$ and $L = \{P_1, P_2, ...P_K, P_C\}$, where each $P_i$ (for $i \in \{1..K\}$) are selected randomly.

**Dataset 2 in I2R - Level 2.** In Level 2, we create a more complex recommendation task. We consider the list $L$ with two distinct halves. The first considers random instances from different disciplines of $P_C$, and the second one samples random instances considering the same discipline as $P_C$. The resulting list is $L = \{P_1, P_2, ...P_K, P_C\}$, where $P_{i\in\{1...\frac{K}{2}\}}$ are random papers from different disciplines than $P_C$, and $P_{j\in\{\frac{K}{2}...K\}}$ are random papers with the same field of $P_C$.

To sum up, in this section we describe the main tasks and dataset configuration of IDRBench. For the tasks, we adopt a progressive evaluation approach. For the dataset, we define a set of configurations to mimic real-world scenarios. We emphasize the importance of constructing datasets in which papers $P_B$ and $P_C$ exhibit a meaningful and reliable distance between their subject disciplines, especially for tasks I3 and I2R. This distinction is crucial for evaluating whether LLMs are truly capable of identifying interdisciplinary research (IDR). For positive cases, we rely on the expertise of our annotators to select paper with reliable distance. While for negative cases, we employ random selection strategies. At difficulty level 1, papers are randomly selected from clearly distinct disciplines. At difficulty level 2, we select paper pairs from the same discipline. This design aims to create counterfactual examples, in which papers belong to the same disciplinary category - making them a feasible combination but not necessarily an instance of IDR. This setup challenges the model to go beyond surface-level cues and assess deeper conceptual relationships.

## 4 Experiment Setup

**Model Setup.** We run the 3 tasks introduced in Section 3.2 with 10 models, including 5 LLMs with reasoning capabilities. The model temperature is set to be 1.0 and maximum output tokens to be 500. We use zero-shot prompting in I2R and we use both zero-shot and few-shot prompting [5, 17, 35] in IPI and I3 ( with 5 instances in IPI and 3 in I3). We curate few-shot learning examples with a mixture of positive and negative samples from our task datasets. We also ensure that the knowledge cutoff date for the models in our experiments has no overlap with the earliest date of the paper that we collected from arXiv. Details of model setup can be found in Appendix A.2.1.

**Task Setup.** We provide two different implementation setups for the three tasks described in Section 3. In IPI and I3, we construct regular expressions to extract LLM answers. For I2R, and following the discussion in [32], considering that pairwise comparison can provide better results than asking directly to predict the final re-ranking decision. We apply the same Swiss tournament used in [32] with the number of rounds to be 10. Detailed prompts are provided in Appendix A.2.3.

**Evaluation Metrics.** For classification task in IPI and I3, we use F1 and Macro-F1 score to report the results. For recommendation task in I2R, we use Mean Reciprocal Rank (MRR) [8] and Normalized Discounted Cumulative Gain (NDCG@10) [14, 34] to report the results. For statistical significance test we have used the Wilcoxon signed-rank [13]. We include the details of evaluation metrics for each task in Appendix A.2.2.

## 5 Evaluation with IDRBench

### 5.1 Results

**IPI Task Performance.** The first two rows in Table 2 report the F1 scores of various LLMs on the IPI task. We compare two different prompting methods, using 0-shot and 5-shot. We conduct our evaluation using 120 positive samples and 1,200 negative samples in our dataset. Compared to the F1 score of 0.176 for the random-guessing baseline, we can see all models achieve higher performance, with an average F1 score of 0.371 under zero-shot setting and 0.364 under few-shot setting, with `llama-3.1-70B` model achieving the highest performance of 0.416. Interestingly, the augmentation with few-shot samples is not always producing positive performance gains, and sometimes even affects F1 negatively, e.g. the F1 scores obtained from `llama-3.1-70B` and `claude-3.7-sonnet`.

| Prompting | Metric | ChatGPT Mini | | | Gemini | | Llama 3 | | Claude | DeepSeek | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | `4o` | `o1` | `o3` | `1.5-pro` | `2.0` | `3.1` | `3.3` | `3.7` | `v3` | `r1` |
| 0-shot | Precision | 0.268 | 0.264 | 0.235 | 0.161 | 0.205 | **0.283** | 0.227 | 0.211 | 0.274 | 0.254 |
| | Recall | 0.758 | 0.942 | 0.975 | **1.000** | 0.950 | 0.778 | 0.850 | 0.967 | 0.658 | 0.950 |
| | F1 | 0.396 | 0.412 | 0.379 | 0.277 | 0.337 | **0.416** | 0.358 | 0.347 | 0.387 | 0.401 |
| 5-shot | Precision | 0.266 | 0.283 | 0.275 | 0.189 | 0.232 | 0.177 | 0.193 | 0.174 | 0.235 | **0.310** |
| | Recall | 0.758 | 0.958 | 0.975 | 0.908 | 0.767 | 0.933 | 0.958 | **0.983** | 0.642 | 0.875 |
| | F1 | 0.394 | 0.436 | 0.429 | 0.313 | 0.357 | 0.298 | 0.321 | 0.295 | 0.344 | **0.458** |

Table 2: Classification accuracy of LLMs on IPI task in IDRBench.

**I3 Task Performance.** We also report I3 in F1 scores to represent classification accuracy in the last four rows in Table 2 using two datasets with different difficulty levels. Compared to a random guessing baseline of 0.176, we observe a higher average F1 score of 0.550 for zero shot and 0.505 for few shot setting, with the `gpt-4o-mini` leading in three out of the four settings, and `llama-3.3` ranked top in the zero shot setting on dataset level 1. We also observe a notable performance decrease when the models are tested on level 2 dataset, with an average drop of 58.8% in zero-shot setting and 65.3% in few-shot setting compared to the performance in level 1 dataset.

| | Prompting | Metric | ChatGPT Mini | | | Gemini | | Llama 3 | | Claude | DeepSeek | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | `4o` | `o1` | `o3` | `1.5-pro` | `2.0` | `3.1` | `3.3` | `3.7` | `v3` | `r1` |
| **Level 1** | 0-shot | Precision | **0.867** | 0.549 | 0.138 | 0.560 | 0.511 | 0.612 | 0.828 | 0.369 | 0.805 | 0.486 |
| | | Recall | 0.542 | 0.467 | **0.875** | 0.467 | 0.750 | 0.607 | 0.642 | 0.400 | 0.583 | 0.725 |
| | | F1 | 0.667 | 0.505 | 0.239 | 0.509 | 0.608 | 0.609 | **0.723** | 0.384 | 0.676 | 0.582 |
| | 3-shot | Precision | **0.820** | 0.478 | 0.135 | 0.210 | 0.282 | 0.382 | 0.528 | 0.547 | 0.642 | 0.258 |
| | | Recall | 0.683 | 0.367 | 0.675 | 0.750 | **0.975** | 0.892 | 0.858 | 0.775 | 0.733 | 0.767 |
| | | F1 | **0.745** | 0.415 | 0.226 | 0.328 | 0.437 | 0.535 | 0.654 | 0.641 | 0.685 | 0.386 |
| **Level 2** | 0-shot | Precision | **0.336** | 0.101 | 0.048 | 0.165 | 0.103 | 0.158 | 0.264 | 0.073 | 0.192 | 0.115 |
| | | Recall | 0.544 | 0.367 | **0.844** | 0.467 | 0.733 | 0.663 | 0.678 | 0.367 | 0.556 | 0.722 |
| | | F1 | **0.415** | 0.158 | 0.091 | 0.243 | 0.181 | 0.255 | 0.380 | 0.122 | 0.286 | 0.198 |
| | 3-shot | Precision | **0.220** | 0.090 | 0.049 | 0.054 | 0.073 | 0.092 | 0.109 | 0.131 | 0.153 | 0.071 |
| | | Recall | 0.700 | 0.333 | 0.733 | 0.700 | **0.989** | 0.833 | 0.867 | 0.733 | 0.722 | 0.789 |
| | | F1 | **0.334** | 0.142 | 0.092 | 0.100 | 0.135 | 0.166 | 0.193 | 0.223 | 0.253 | 0.130 |

Table 3: Classification accuracy of LLMs on I3 task in IDRBench.

**I2R Task Performance.** Table 4 presents the I2R recommendation results in IDRBench. We apply the Mean Reciprocal Rank (MRR) metric that emphasizes the ranking accuracy by rewarding correct items ranked higher. We note that similar ranking metrics (e.g., NDCG@10) also find similar results. To ensure the reliability of our results, we also assess the statistical significance test when compared to the best results (highlighted in bold) using the Wilcoxon signed-rank test [13] – marked in asterisks in the table. For both datasets Level 1 and Level 2, `gemini-1.5-pro` achieves the highest absolute score of 0.703 and 0.662, respectively. However, the performance of different models, as measured by the Wilcoxon test, shows less variance within each difficulty level, indicating similar recommendation abilities for the models.

| Difficulty | Metric | ChatGPT Mini | | | Gemini | | Llama 3 | | Claude | DeepSeek | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | `4o` | `o1` | `o3` | `1.5-pro` | `2.0` | `3.1` | `3.3` | `3.7` | `v3` | `r1` |
| **Level 1** | MRR | 0.682* | 0.589 | 0.640 | **0.703** | 0.687* | 0.622 | 0.619 | 0.679* | 0.674* | 0.588 |
| **Level 2** | | 0.645* | 0.500 | 0.571 | **0.662** | 0.621* | 0.576 | 0.574 | 0.636* | 0.650* | 0.526 |

Table 4: I2R Results – "$*$" indicates that there is no statistical significance against the bold value.

### 5.2 Detailed Analysis

*Are LLMs capable of conducting interdisciplinary research?* Our results indicate mixed answers: there are certain tasks where LLMs show promising performance, such as I3 and I2. However, in IPI, LLMs still require improvement. To provide more comprehensive answers as well as insights to this question, we conduct further analysis on each task to reveal the strengths and limitations of these benchmarking models.

**LLMs are better at identifying disciplines in IDR.** In addition to the classification accuracy in IPI described in the previous section, one might wonder whether LLMs can also correctly identify the set of disciplines that the citing paper $P_A$ is integrating. We present the performance of different models in Table 5 and report discipline classification accuracy under the zero-shot setting. We observe that although LLMs demonstrate lower accuracy in IDR paper identification with an average F1 score around 0.371 (0-shot line in Table 2), they achieve high performance in identifying the disciplines to which $P_A$ belongs.

| ChatGPT Mini | | | Gemini | | Llama 3 | | Claude | DeepSeek | |
|---|---|---|---|---|---|---|---|---|---|
| `4o` | `o1` | `o3` | `1.5-pro` | `2.0` | `3.1` | `3.3` | `3.7` | `v3` | `r1` |
| 0.845 | 0.877 | 0.874 | 0.906 | 0.903 | 0.795 | 0.898 | 0.876 | 0.895 | 0.884 |

Table 5: 0-shot discipline identification accuracy among positive samples, reported in F1-Score.

**LLMs still struggle to generate quality IDR ideas.** In addition to the I3 classification results, we further investigate the quality of the IDR ideas among the positive answers that LLMs provided in I3 task. We perform two comparisons using the Output Idea $I_O$, including i) Integration reason, which compares LLM output ("Integration Reason", Figure 2), with annotated gold labels [6] on how the cited papers $(P_B, P_C)$ are integrated in $P_A$, and ii) Abstract, which compares the actual abstract of IDR paper $P_A$ with the abstract generated by LLM ("Abstract", Figure 2). The quality metrics are two-folded: semantic similarity (measured by SciBERT Similarity [2]) and content similarity (measured by BLEU[28] and ROUGE-1[19]). Table 6 presents the comparison results of IDR idea quality across different LLMs. Interestingly, both reasoning and abstract generation exhibit a strong alignment with the annotated samples in semantic similarity, achieving an average similarity of 0.756 and 0.812. However, in terms of content similarity, no model demonstrates a high match with the original text. This divergence suggests that LLMs are focusing more on the semantics of IDR ideation rather than the content itself, indicating substantial potential for improvement in automating the idea generation process in IDR. We provide a specific case study of the LLM reasoning in Appendix A.3.

| | **Metric** | ChatGPT Mini | | | Gemini | | Llama 3 | | Claude | DeepSeek | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | `4o` | `o1` | `o3` | `1.5-pro` | `2.0` | `3.1` | `3.3` | `3.7` | `v3` | `r1` |
| **Integration Reason** | Similarity | 0.747 | 0.759 | **0.785** | 0.744 | 0.745 | 0.756 | 0.763 | 0.747 | 0.760 | 0.757 |
| | BLEU | 0.013 | 0.010 | 0.011 | 0.012 | 0.012 | **0.014** | 0.006 | 0.011 | 0.011 | 0.011 |
| | ROUGE-1 | **0.218** | 0.160 | 0.182 | 0.186 | 0.206 | 0.215 | 0.145 | 0.195 | 0.192 | 0.161 |
| **Abstract** | Similarity | 0.800 | 0.830 | 0.818 | 0.802 | 0.798 | 0.793 | 0.785 | 0.829 | 0.831 | **0.837** |
| | BLEU | 0.015 | 0.014 | 0.014 | 0.016 | 0.016 | 0.017 | **0.018** | 0.014 | 0.014 | 0.012 |
| | ROUGE-1 | 0.328 | 0.321 | 0.321 | 0.325 | 0.331 | 0.328 | **0.338** | 0.314 | 0.318 | 0.307 |

Table 6: Similarity comparison of integration reason and abstract between LLM-generated responses and the actual IDR paper. We use I3-Level 1 dataset and 0-shot setting.

**LLMs are self-conscious IDR idea recommenders.** Given the recommendation results in I2R task, we also investigate whether the ranking results are influenced by the semantic similarity between

[6] In our annotation process, annotators selected the most important sentences from the abstract that describe the interdisciplinary integration.

cited paper $P_B$ and each candidate in the paper list $L$, rather than the IDR idea combination. In other words, we analyze whether LLMs have developed an awareness that helps them distinguish IDR integration from paper similarity. Table 7 reports the Kendall's $\tau$ correlation between the rankings produced by the LLM and those based on contextual similarity (using SciBERT similarity). As shown in the table, the correlation ranges from approximately 0.1 to 0.2, indicating that contextual similarity does not strongly influence the LLM's re-ranking decisions, further suggesting that LLMs are capable of handling basic IDR recommendation tasks without reliance on paper similarity.

| **Difficulty** | **ChatGPT Mini** | | | **Gemini** | | **Llama 3** | | **Claude** | **DeepSeek** | |
|---|---|---|---|---|---|---|---|---|---|---|
| | `4o` | `o1` | `o3` | `1.5-pro` | `2.0` | `3.1` | `3.3` | `3.7` | `v3` | `r1` |
| **Level 1** | 0.196 | 0.199 | 0.180 | 0.233 | 0.220 | 0.221 | 0.211 | 0.249 | 0.232 | 0.194 |
| **Level 2** | 0.180 | 0.158 | 0.162 | 0.196 | 0.184 | 0.184 | 0.187 | 0.200 | 0.194 | 0.169 |

Table 7: Kendall correlation among I2R ranking and similarity scores of seed and candidate papers.

## 6 Conclusion

We introduce IDRBench, a novel benchmark to probe LLMs' capacity for interdisciplinary research. IDRBench includes a high quality dataset collected from human annotation. We also design a set of three progressive tasks: *(i) IDR Paper Identification (IPI)*, *(ii) IDR Idea Integration (I3)*, and *(iii) IDR Idea Recommendation (I2R)* — aiming to answer our research question: *Are LLMs capable of conducting interdisciplinary research?* Throughout experiment result and further analysis, we uncover that LLMs have already fostered some level of IDR awareness, but they still struggle to produce quality IDR ideas in a plug-and-play fashion. To the best of our knowledge, IDRBench is the first to rigorously investigate LLMs' IDR ability and to provide insights that encourage follow-up studies. We hope IDRBench can become the "Eureka" moment for upcoming studies that inspire the development of more benchmarks to evaluate LLM's IDR ability.

## 7 Limitations

Our benchmark evaluates the inherent ability of LLMs to conduct interdisciplinary research using the title and abstract of scientific papers as input. Although these elements are widely accepted in the literature as sufficiently informative, incorporating full paper content could offer a richer and more comprehensive perspective. While ArXiv provides a reliable foundation for evaluating IDR with LLMs, expanding to the broader range of disciplines would further fortify the relevance and generalizability of our findings.

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# A Appendix

## A.1 Dataset Details

This section describes the details of our IDRBench dataset, including the details of our custom web platform and annotator background details.

### A.1.1 Annotation Platform

To collect the positive samples in IDRBench, we designed a custom Web platform that includes four tasks for the annotators to complete. Specifically, the annotators are first provided with a panel of papers shown in Figure 4 when logged in, where they can choose the paper that they feel comfortable annotating. In task one, shown in Figure 5, they are instructed to first read the title and abstract of a given paper and decide whether it is an IDR paper. If so, they are directed to task two (Figure 6) to indicate the research type (i.e., whether it is basic research or applied research) of the IDR paper. In the third task, the annotators are asked to specify the exact papers that contribute to the IDR idea in this paper. They can add up to 4 papers that describe the IDR idea, as shown in the screenshot of Figure 7. Finally, they are asked to annotate the specific sentence(s) in this IDR paper that specifically describe such integration, as shown in Figure 8. The Web platform is also designed in a way where the annotators can revert any progress they have made so far in case they changed their minds, shown in Figure 9.

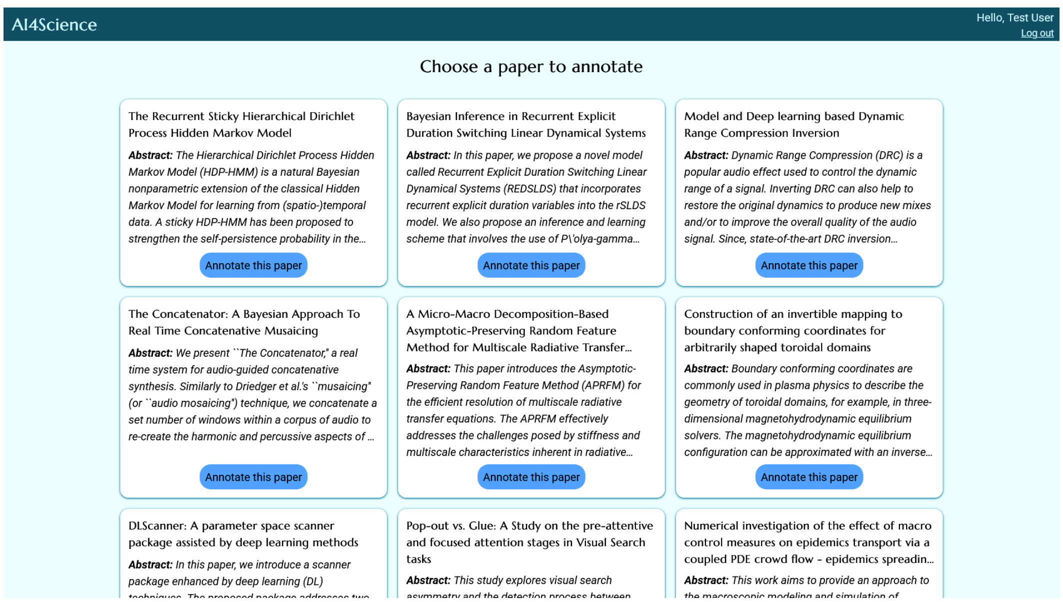


Figure 4: List of papers available for the annotator to choose from.

### A.1.2 Human Annotation

We recruit 8 students from diverse academic backgrounds to perform the human annotation. Each student received about \$18 for each work hour. The academic level of the annotators ranges from the fourth year of undergraduate studies to the second year of PhD. Table 8 showcases a summary of the annotators' background, including their level of study and their area of expertise.

## A.2 Experiment Details

This section introduces the details of our experiment setup using IDRBench. Section A.2.1 introduces the models that we use in our evaluation, Section A.2.2 lists the evaluation metrics used to quantify the performance of different models, and Section A.2.3 provides the prompts that we used in our experiments.

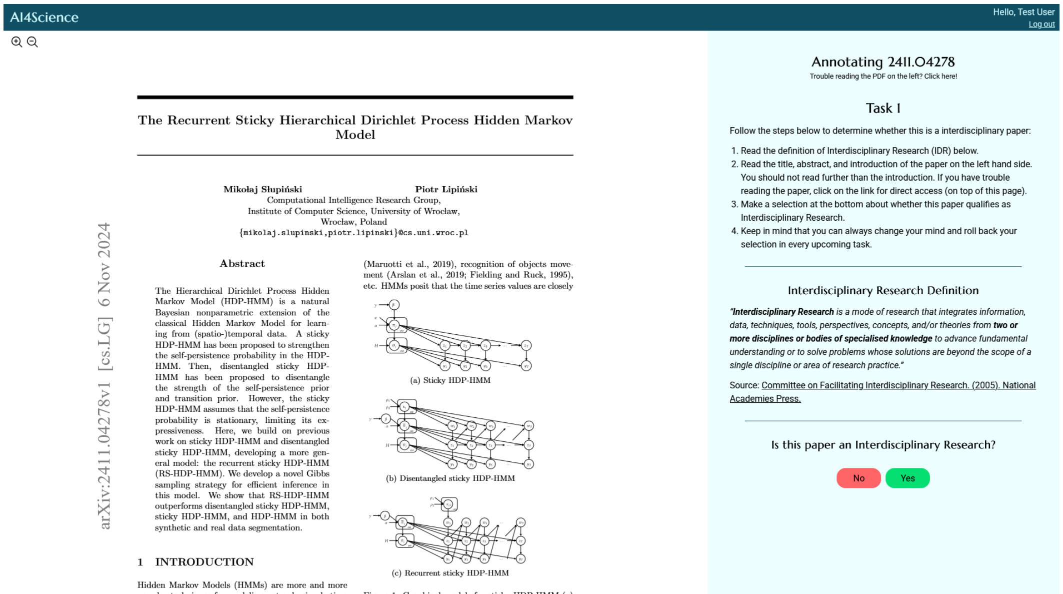


Figure 5: Task 1, displayed after the annotator selects a paper.

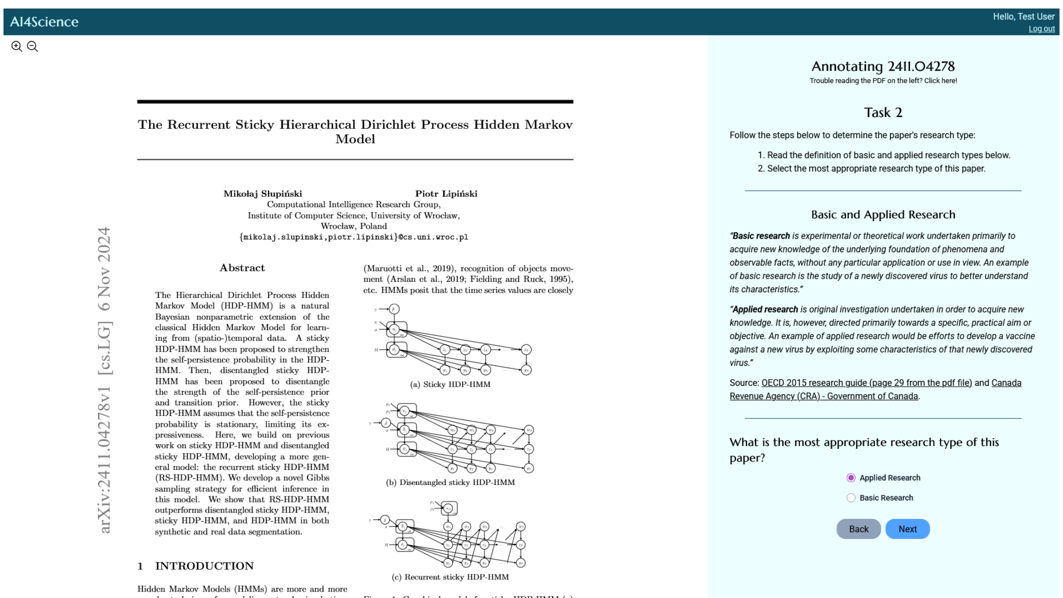


Figure 6: Task 2, displayed if the annotator answers "Yes" in Task 1.

<table>
<tr><th>Level of study</th><th>Count</th><th>Expertise Area</th></tr>
<tr><td>Undergraduate</td><td>3</td><td rowspan="3">Robotics, Control, AI, Natural Language Processing,<br>Robotics, Rehabilitation, AI, Medical science,<br>Mechanical and Materials Engineering, Mechatronics and Robotics Engineering,<br>Brain-computer interfaces, Machine Learning</td></tr>
<tr><td>Master’s</td><td>2</td></tr>
<tr><td>PhD</td><td>3</td></tr>
</table>

Table 8: Academic background summary of annotators.


Figure 7: Task 3, displayed after completing Task 2.


Figure 8: Task 4, displayed after completing Task 3.

### A.2.1 Models

**ChatGPT.** First released in November 2022, `ChatGPT` is a series of models that is trained by OpenAI to conduct complex tasks using natural language prompts [27, 26]. In our paper, we select two types of GPT models from the GPT family, where `gpt-4o-mini` focuses on instruction following abilities, and `gpt-o1-mini` and `gpt-o3-mini` have enhanced reasoning capabilities in complex tasks.

**Claude-3.7.** This is the latest model released by Anthropic that incorporates an adaptive thinking process when prompted with a task. Specifically, it simulates an adaptive switch between the system 1 thinking and system 2 thinking, akin to human [1]. We use the latest model `claude-3.7-sonnet` to run our benchmark.

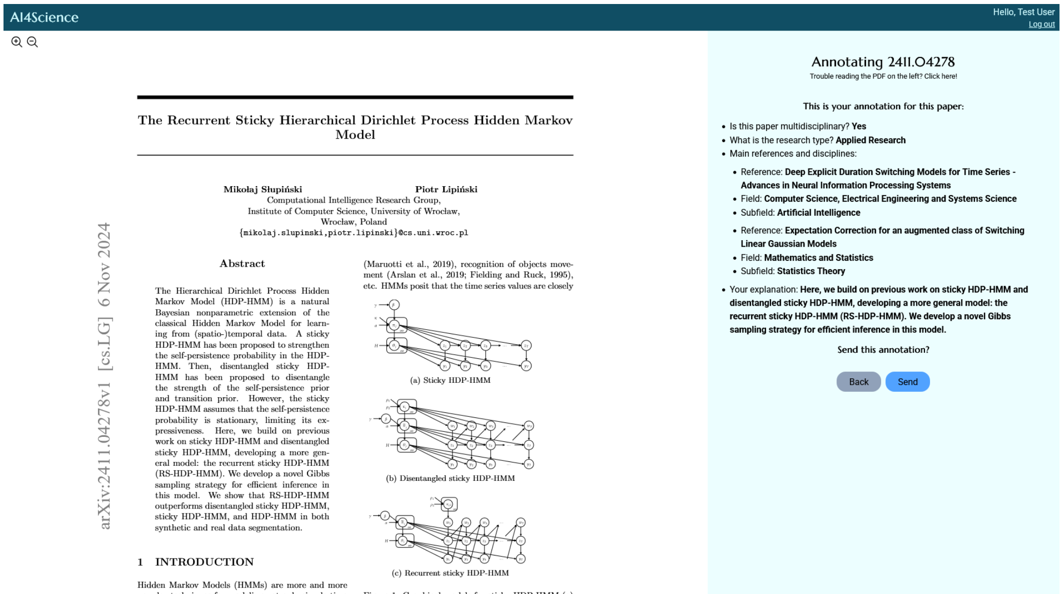


Figure 9: Review page, where the annotator can review and optionally go back and edit their answers.

**Llama-3.** As an updated version from `llama-2` [11], `llama-3` is trained with more recent corpora from various sources and achieves a better performance in various benchmarks. Different from the GPT family, Llama models are completely open-source. In our evaluation, we use `llama-3.1-70b` and `llama-3.3-70b`, the two most powerful models in the family so far.

**Deepseek.** Deepseek is a suite of models trained under a schema that has higher efficiency yet with a comparable performance to those closed-source models [9]. The Deepseek family includes a basic chat model called `deepseek-v3` and an advanced reasoning model `deepseek-r1`. We include both the chat model and the reasoning model in the results.

| **Model** | **Cutoff Date** | **Release Date** |
|---|---|---|
| `gpt-4o-mini` | Oct 2023 | Jul 2024 |
| `gpt-o1-mini` | Oct 2023 | Jul 2024 |
| `gpt-o3-mini` | Oct 2023 | Jul 2024 |
| `llama-3.1-70B` | Dec 2023 | Jul 2024 |
| `llama-3.3-70B` | Dec 2023 | Jul 2024 |
| `gemini 1.5-pro` | May 2024 | May 2024 |
| `gemini-2.0-flash` | Jun 2024 | Dec 2024 |
| `claude-3.7-Sonnet` | Nov 2024 | Feb 2025 |
| `deepSeek-v3` | Jul 2024 | Dec 2024 |
| `deepSeek-r1` | Oct 2023 | Dec 2024 |
| **Earliest paper in IDRBench** | Nov 2024 | |

Table 9: Knowledge cutoff dates of LLMs' pretraining data.

### A.2.2 Evaluation Metrics

We use a set of mainstream metrics for both our classification tasks and recommendation tasks. For the classification tasks, we report F1 and Macro F1 scores, and for the recommendation tasks, we report MRR. MRR measures, on average, how far down the ranked list the first relevant item appears. Specifically, for a set of queries $Q$, let $\mathrm{rank}_q$ be the position of the first relevant item for query $q \in Q$.

| IPI prompt |
| --- |
| Read the title and abstract of a given academic paper and identify whether this is an interdisciplinary research paper. Also, select one or more subjects from the list below to indicate which subject(s) does this paper belong to. After you provide your verdict and your choice, provide a score from 0 to 100 to indicate your confidence level in the correctness of the verdict.<br>The official definition of a typical interdisciplinary paper can be found below:<br>Interdisciplinary Research is a mode of research that integrates information, data, techniques, tools, perspectives, concepts, and/or theories from two or more disciplines or bodies of specialised knowledge to advance fundamental understanding or to solve problems whose solutions are beyond the scope of a single discipline or area of research practice.<br>Think carefully to make your verdict, answer "Yes" when this is a valid IDR paper. Otherwise, answer "No".<br>Note: The confidence level indicates the degree of certainty you have about your verdict and is represented as a percentage. For instance, if your confidence level is 80, it means you are 80 percent certain that your answer is correct and there is a 20 percent chance that it may be incorrect.<br><br>Paper title: %s;<br>Paper abstract: %s;<br><br>Subject list: ["Computer Science, Electrical Engineering and System Science", "Economics and Quantitative Finance", "Mathematics and Statistics", "Physics", "Quantitative Biology", "Other"]<br><br>Use the template (in this format, with no markdown and lines separated by '\n') below to provide your answer.<br>Your verdict: A simple answer containing either "Yes" or "No".<br>Confidence score: A numeric score ranging from 0 to 100<br>Subject: Your choice of subjects from the list above. Use a list with square brackets "[]" separated by comma and remember to use "" to wrap your answer. |

Table 10: Prompt used for the IPI task.

The Reciprocal Rank (RR) for $q$ is

$$\mathrm{RR}_q = \begin{cases} \dfrac{1}{\mathrm{rank}_q}, & \text{if a relevant item is found,} \\ 0, & \text{otherwise.} \end{cases}$$

The Mean Reciprocal Rank is then

$$\mathrm{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{RR}_q = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\mathrm{rank}_q}.$$

#### A.2.3 Prompts

In this section, we list the prompts that we used in our evaluation, including 1) IDR Paper Identification (IPI), 2) IDR Idea Integration (I3), and 3) IDR Idea Recommendation (I2R).

#### A.2.4 Experiment Cost

We use a variety of API-based platforms that host the models in our evaluation. The cost of the experiment was calculated based on the per-million input/output token rates listed by each host platform.

### A.3 Analysis

In addition to the reporting of similarity scores in Section 5, we also conduct a qualitative analysis on the LLM output reasons against human researchers' original writing on formulating a potential

| I3 prompt |
| --- |
| Read the title and abstract of papers from two disciplines and decide whether you can extract concepts from both disciplines to create a novel multidisciplinary research idea. After you provide your verdict, provide a score from 0 to 100 to indicate your confidence level in the correctness of the verdict.<br>Keep in mind a good Interdisciplinary Research idea includes the following standards:<br>* This research idea should be Interdisciplinary, whereas the idea stems from the combination of ideas from the two papers introduced above.<br>* The Interdisciplinary Research ideas should follow this definition: Interdisciplinary Research is a mode of research that integrates information, data, techniques, tools, perspectives, concepts, and/or theories from two or more disciplines or bodies of specialised knowledge to advance fundamental understanding or to solve problems whose solutions are beyond the scope of a single discipline or area of research practice.<br>* This research idea should be feasible, whereas the hypothesis is not purely theoretical and can be validated by experiments.<br>* This research idea should be novel, whereas it is not only rare but also ingenious, imaginative, or surprising.<br>* This research idea should be useful, whereas it applies to the stated problem and is effective at solving the problem.<br>Think carefully to make your decision, and you should only answer "Yes" when this multidisciplinary idea meets ALL of the standards above. Otherwise, you should answer "No".<br>Note: The confidence level indicates the degree of certainty you have about your verdict and is represented as a percentage. For instance, if your confidence level is 80, it means you are 80 percent certain that your answer is correct and there is a 20 percent chance that it may be incorrect.<br><br>Paper in Discipline 1:<br>%s<br><br>Paper in Discipline 2:<br>%s<br><br>Use the template (in this format, with no markdown and lines separated by '\n') to provide your answer.<br>Your verdict: A simple answer containing either "Yes" or "No".<br>Your reason: A short paragraph less than 50 words briefly describes your reasons that you made the verdict above.<br>Confidence score: A numeric score ranging from 0 to 100 |

Table 11: Prompt used for the I3 task.

IDR research. Our analysis is twofold: on one hand, we compare the key integration reasoning from LLM output with the annotated sentences marked by the annotators; on the other hand, we compare the full abstract generated by LLM with the abstract of the targeted IDR paper. Figure 10 provides a comparison between the annotation and the LLM-generated response. As we can see, although LLMs like `llama-3.3-70b` can generate topic-related content in both integration reasoning and abstract generation, they still lack detailed methods and a solid experiment schedule in formulating an IDR project.

| I2R prompt (0-shot) |
| --- |
| In this task, you are given a main paper introducing the key concepts that provides certain parts in a Interdisciplinary idea as well as two candidate papers that forms the remaining parts of a Interdisciplinary idea. Compare them and select which one is better to pair with the main paper in forming a multidisciplinary idea. After you provide your selection, provide a score from 0 to 100 to indicate your confidence level in the correctness of making this choice.<br>Keep in mind a good Interdisciplinary Research idea includes the following standards:<br>* This research idea should be Interdisciplinary, whereas the idea stems from the combination of ideas from the two papers introduced above.<br>* The Interdisciplinary Research ideas should follow this definition: Interdisciplinary Research is a mode of research that integrates information, data, techniques, tools, perspectives, concepts, and/or theories from two or more disciplines or bodies of specialised knowledge to advance fundamental understanding or to solve problems whose solutions are beyond the scope of a single discipline or area of research practice.<br>* This research idea should be feasible, whereas the hypothesis is not purely theoretical and can be validated by experiments.<br>* This research idea should be novel, whereas it is not only rare but also ingenious, imaginative, or surprising.<br>* This research idea should be useful, whereas it applies to the stated problem and is effective at solving the problem.<br>Note: The confidence level indicates the degree of certainty you have about your verdict and is represented as a percentage. For instance, if your confidence level is 80, it means you are 80 percent certain that your answer is correct and there is a 20 percent chance that it may be incorrect.<br><br>Main paper title: %s;<br>Main paper abstract: %s;<br><br>Paper 1 title: %s;<br>Paper 1 abstract: %s;<br><br>Paper 2 title: %s;<br>Paper 2 abstract: %s;<br><br>Use the template (in this format, with no markdown and lines separated by '\n') to provide your answer.<br>Your choice: A simple answer containing either "Paper 1" or "Paper 2".<br>Confidence score: A numeric score ranging from 0 to 100 |

Table 12: Prompt used for the I2R task.

| IPI Few-shot Examples |
|---|
| Example 1:<br>Paper title: Designing a Light-based Communication System with a Biomolecular Receiver; Paper abstract: Biological systems transduce signals from their surroundings in numerous ways. This paper introduces a communication system using the light-gated ion channel Channelrhodopsin-2 (ChR2), which causes an ion current to flow in response to light. Our design includes a ChR2-based receiver along with encoding, modulation techniques and detection. Analyzing the resulting communication system, we discuss the effect of different parameters on the performance of the system. Finally, we discuss its potential design in the context of bio-engineering and light-based communication and show that the data rate scales up with the number of receptors, indicating that high-speed communication may be possible.<br>Your verdict: Yes<br><br>Example 2:<br>Paper title: BarcodeMamba: State Space Models for Biodiversity Analysis; Paper abstract: DNA barcodes are crucial in biodiversity analysis for building automatic identification systems that recognize known species and discover unseen species. Unlike human genome modeling, barcode-based invertebrate identification poses challenges in the vast diversity of species and taxonomic complexity. Among Transformer-based foundation models, BarcodeBERT excelled in species-level identification of invertebrates, highlighting the effectiveness of self-supervised pretraining on barcode-specific datasets. Recently, structured state space models (SSMs) have emerged, with a time complexity that scales sub-quadratically with the context length. SSMs provide an efficient parameterization of sequence modeling relative to attention-based architectures. Given the success of Mamba and Mamba-2 in natural language, we designed BarcodeMamba, a performant and efficient foundation model for DNA barcodes in biodiversity analysis. We conducted a comprehensive ablation study on the impacts of self-supervised training and tokenization methods, and compared both versions of Mamba layers in terms of expressiveness and their capacity to identify "unseen" species held back from training. Our study shows that BarcodeMamba has better performance than BarcodeBERT even when using only 8.3% as many parameters, and improves accuracy to 99.2% on species-level accuracy in linear probing without fine-tuning for "seen" species. In our scaling study, BarcodeMamba with 63.6% of BarcodeBERT's parameters achieved 70.2% genus-level accuracy in 1-nearest neighbor (1-NN) probing for unseen species.;<br>Your verdict: Yes<br><br>Example 3:<br>Paper title: An ADHD Diagnostic Interface Based on EEG Spectrograms and Deep Learning Techniques;<br>Paper abstract: This paper introduces an innovative approach to Attention-deficit/hyperactivity disorder (ADHD) diagnosis by employing deep learning (DL) techniques on electroencephalography (EEG) signals. This method addresses the limitations of current behavior-based diagnostic methods, which often lead to misdiagnosis and gender bias. By utilizing a publicly available EEG dataset and converting the signals into spectrograms, a Resnet-18 convolutional neural network (CNN) architecture was used to extract features for ADHD classification. The model achieved a high precision, recall, and an overall F1 score of 0.9. Feature extraction highlighted significant brain regions (frontopolar, parietal, and occipital lobes) associated with ADHD. These insights guided the creation of a three-part digital diagnostic system, facilitating cost-effective and accessible ADHD screening, especially in school environments. This system enables earlier and more accurate identification of students at risk for ADHD, providing timely support to enhance their developmental outcomes. This study showcases the potential of integrating EEG analysis with DL to enhance ADHD diagnostics, presenting a viable alternative to traditional methods.;<br>Your verdict: Yes |

Table 13: Few-shot learning samples in IPI task.

| **IPI Few-shot Examples (Cont.)** |
|---|
| Example 4:<br>Paper title: Graph Neural Controlled Differential Equations For Collaborative Filtering; Paper abstract: Graph Convolution Networks (GCNs) are widely considered state-of-the-art for recommendation systems. Several studies in the field of recommendation systems have attempted to apply collaborative filtering (CF) into the Neural ODE framework. These studies follow the same idea as LightGCN, which removes the weight matrix or with a discrete weight matrix. However, we argue that weight control is critical for neural ODE-based methods. The importance of weight in creating tailored graph convolution for each node is crucial, and employing a fixed/discrete weight means it cannot adjust over time within the ODE function. This rigidity in the graph convolution reduces its adaptability, consequently hindering the performance of recommendations. In this study, to create an optimal control for Neural ODE-based recommendation, we introduce a new method called Graph Neural Controlled Differential Equations for Collaborative Filtering (CDE-CF). Our method improves the performance of the Graph ODE-based method by incorporating weight control in a continuous manner. To evaluate our approach, we conducted experiments on various datasets. The results show that our method surpasses competing baselines, including GCNs-based models and state-of-the-art Graph ODE-based methods.;<br>Your verdict: No<br><br>Example 5:<br>Paper title: Mechano-Bactericidal Surfaces Achieved by Epitaxial Growth of Metal-Organic Frameworks;<br>Paper abstract: Mechano-bactericidal (MB) surfaces have been proposed as an emerging strategy for preventing biofilm formation. Unlike antibiotics and metal ions that chemically interfere with cellular processes, MB nanostructures cause physical damage to the bacteria. The antibacterial performance of artificial MB surfaces relies on rational control of surface features, which is difficult to achieve for large surfaces in real-life applications. Herein, we report a facile and scalable method for fabricating MB surfaces based on metal-organic frameworks (MOFs) using epitaxial MOF-on-MOF hybrids as building blocks with nanopillars of less than 5 nm tip diameter, 200 nm base diameter, and 300 nm length. Two methods of MOF surface assembly, in-situ growth and ex-situ dropcasting, result in surfaces with nanopillars in different orientations, both presenting MB actions (bactericidal efficiency of 83% for E. coli). Distinct MB mechanisms, including stretching, impaling, and apoptosis-like death induced by mechanical injury are discussed with the observed bacterial morphology on the obtained MOF surfaces.;<br>Your verdict: No |

Table 14: Few-shot learning samples in IPI task (Cont.).

| I3 Few-shot Examples |
|---|
| Example 1:<br>Paper in Discipline 1:<br>- title: "Relation Between Retinal Vasculature and Retinal Thickness in Macular Edema"<br>- abstract: "This study has investigated the relationship of retinal vasculature and thickness for Macular Edema (ME) subjects. Ninety sets Fluorescein Angiograph (FA) Optical Coherence Tomography (OCT) 54 participants were analyzed. Multivariate analysis using binary logistic regression model was used to association between vessel parameters thickness. The results reveal feature i.e. fractal dimension (FD) as most sensitive parameter changes in associated with ME. Thus, indicating a direct which is caused due neovascular causing exudates, leakages hemorrhages, applications alternate modality detection"<br><br>Paper in Discipline 2:<br>- title: "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale"<br>- abstract: "While the Transformer architecture has become de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used replace certain components of networks while keeping their overall structure place. We show that this reliance on CNNs not necessary and a pure transformer directly sequences image patches can perform very well classification tasks. When pre-trained large amounts data transferred multiple mid-sized small recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision (ViT) attains excellent results compared state-of-the-art requiring substantially fewer computational resources train."<br><br>Your verdict: Yes<br>Your reason: A novel work can combine transformers with two distinct methods that evaluate the quality of retinopathy",<br>Confidence score: 92<br><br>Example 2:<br>Paper in Discipline 1:<br>- title: "Channelrhodopsin-2, a directly light-gated cation-selective membrane channel"<br>- abstract: "Microbial-type rhodopsins are found in archaea, prokaryotes, and eukaryotes. Some of them represent membrane ion transport proteins such as bacteriorhodopsin, a light-driven proton pump, or channelrhodopsin-1 (ChR1), recently identified light-gated channel from the green alga Chlamydomonas reinhardtii . ChR1 ChR2, related microbial-type rhodopsin C. , were shown to be involved generation photocurrents this alga. We demonstrate by functional expression, both oocytes Xenopus laevis mammalian cells, that ChR2 is directly light-switched cation-selective channel. This opens rapidly after absorption photon generate large permeability for monovalent divalent cations. desensitizes continuous light smaller steady-state conductance. Recovery desensitization accelerated extracellular H + negative potential, whereas closing decelerated intracellular expressed mainly under low-light conditions, suggesting involvement photoreception dark-adapted cells. The predicted seven-transmembrane helices characteristic G protein-coupled receptors but reflect different motif Finally, we may used depolarize small simply illumination."<br><br>Paper in Discipline 2:<br>- title: "Shannon capacity of signal transduction for multiple independent receptors, DESIGN AND IMPLEMENTATION OF VISIBLE LIGHT COMMUNICATION SYSTEM IN INDOOR ENVIRONMENT"<br>- abstract: "Cyclic adenosine monophosphate (cAMP) is considered a model system for signal transduction, the mechanism by which cells exchange chemical messages. Our previous work calculated Shannon capacity of single cAMP receptor; however, typical cell may have thousands receptors operating in parallel. In this paper, we calculate transduction with an arbitrary number independent, indistinguishable receptors. By leveraging prior results on feedback receptor, show (somewhat unexpectedly) that achieved IID input distribution, and n times receptor. Visible Light communication (VLC) using White Light Emitting Diode (LED) is a promising technology for next generation communication for short range, high speed wireless data transmission." |

Table 15: Few-shot learning samples in I3 task.

| I3 Few-shot Examples (Cont.) |
| --- |

Your verdict: Yes
Your reason: An interdisciplinary paper can aim to use channelrhodopsin-2 (ChR2), a biomolecule, as a receiver to design a light-based communication system, which is a work related to engineering.
Confidence score: 85

Example 3:
Paper in Discipline 1:
- title: "A General Adaptive Dual-level Weighting Mechanism for Remote Sensing Pansharpening"
- abstract: "Currently, deep learning-based methods for remote sensing pansharpening have advanced rapidly. However, many existing methods struggle to fully leverage feature heterogeneity and redundancy, thereby limiting their effectiveness. We use the covariance matrix to model the feature heterogeneity and redundancy and propose Correlation-Aware Covariance Weighting (CACW) to adjust them. CACW captures these correlations through the covariance matrix, which is then processed by a nonlinear function to generate weights for adjustment. Building upon CACW, we introduce a general adaptive dual-level weighting mechanism (ADWM) to address these challenges from two key perspectives, enhancing a wide range of existing deep-learning methods. First, Intra-Feature Weighting (IFW) evaluates correlations among channels within each feature to reduce redundancy and enhance unique information. Second, Cross-Feature Weighting (CFW) adjusts contributions across layers based on inter-layer correlations, refining the final output. Extensive experiments demonstrate the superior performance of ADWM compared to recent state-of-the-art (SOTA) methods. Furthermore, we validate the effectiveness of our approach through generality experiments, redundancy visualization, comparison experiments, key variables and complexity analysis, and ablation studies. Our code is available at https:github.comJie-1203ADWM."

Paper in Discipline 2:
- title: "Secure Semantic Communication With Homomorphic Encryption"
- abstract: "In recent years, Semantic Communication (SemCom), which aims to achieve efficient and reliable transmission of meaning between agents, has garnered significant attention from both academia and industry. To ensure the security of communication systems, encryption techniques are employed to safeguard confidentiality and integrity. However, traditional cryptography-based encryption algorithms encounter obstacles when applied to SemCom. Motivated by this, this paper explores the feasibility of applying homomorphic encryption to SemCom. Initially, we review the encryption algorithms utilized in mobile communication systems and analyze the challenges associated with their application to SemCom. Subsequently, we employ scale-invariant feature transform to demonstrate that semantic features can be preserved in homomorphic encrypted ciphertext. Based on this finding, we propose a task-oriented SemCom scheme secured through homomorphic encryption. We design the privacy preserved deep joint source-channel coding (JSCC) encoder and decoder, and the frequency of key updates can be adjusted according to service requirements without compromising transmission performance. Simulation results validate that, when compared to plaintext images, the proposed scheme can achieve almost the same classification accuracy performance when dealing with homomorphic ciphertext images. Furthermore, we provide potential future research directions for homomorphic encrypted SemCom."

Your verdict: No
Your reason: The two papers are not related to each other. The first paper focuses on remote sensing pansharpening, while the second paper discusses secure semantic communication with homomorphic encryption. There is no clear interdisciplinary connection between them.
Confidence score: 90

Table 16: Few-shot learning samples in I3 task (Cont.).

**Key Idea Integration Comparison**

**Paper Title:** A lightweight Convolutional Neural Network based on U shape structure and Attention Mechanism for Anterior Mediastinum Segmentation

**LLM I3 Reasoning:** Combining AI-powered lesion segmentation with lung cancer screening guidelines can improve diagnosis accuracy.

**Annotated I3 Reasoning:** In this paper, we introduce a U-shaped structure network to segment AM. Two attention mechanisms were used for maintaining long-range dependencies and localization. In order to have the potential of Multi-Head Self-Attention (MHSA) and a lightweight network, we designed a parallel MHSA named Wide-MHSA (W-MHSA).

**Full Abstract Generation Comparison**

**LLM Abstract:** This study proposes an interdisciplinary approach to investigating incidental anterior mediastinal nodular lesions on chest computed tomography (CT) scans in asymptomatic subjects, with a focus on early detection of lung cancer and thymic epithelial tumors. By integrating a UNet-based neural network with attention mechanisms for automatic detection of anterior mediastinal nodular lesions, and guidelines for lung cancer screening from the American Cancer Society, this research aims to develop a novel computer-aided detection system. The system will analyze low-dose chest CT scans to identify high-risk individuals and detect incidental nodular lesions, which can be potential thymic epithelial tumors or lung cancer. The expected outcome of this study is to improve the accuracy of early detection and diagnosis of lung cancer and thymic epithelial tumors, and to provide valuable insights into the prevalence and characteristics of incidental nodular lesions in asymptomatic subjects.

**True Abstract:** To automatically detect Anterior Mediastinum Lesions (AMLs) in the Anterior Mediastinum (AM), the primary requirement will be an automatic segmentation model specifically designed for the AM. The prevalence of AML is extremely low, making it challenging to conduct screening research similar to lung cancer screening. Retrospectively reviewing chest CT scans over a specific period to investigate the prevalence of AML requires substantial time. Therefore, developing an Artificial Intelligence (AI) model to find location of AM helps radiologist to enhance their ability to manage workloads and improve diagnostic accuracy for AMLs. In this paper, we introduce a U-shaped structure network to segment AM. Two attention mechanisms were used for maintaining long-range dependencies and localization. In order to have the potential of Multi-Head Self-Attention (MHSA) and a lightweight network, we designed a parallel MHSA named Wide-MHSA (W-MHSA). Maintaining long-range dependencies is crucial for segmentation when we upsample feature maps. Therefore, we designed a Dilated Depth-Wise Parallel Path connection (DDWPP) for this purpose. In order to design a lightweight architecture, we introduced an expanding convolution block and combine it with the proposed W-MHSA for feature extraction in the encoder part of the proposed U-shaped network. The proposed network was trained on 2775 AM cases, which obtained an average Dice Similarity Coefficient (DSC) of 87.83%, mean Intersection over Union (IoU) of 79.16%, and Sensitivity of 89.60%. Our proposed architecture exhibited superior segmentation performance compared to the most advanced segmentation networks, such as Trans Unet, Attention Unet, Res Unet, and Res Unet++.

Figure 10: Qualitative comparison sample on both idea integration reasoning and full abstract generation using `llama-3.3-70b`, both under zero-shot setting.